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[22CSE279]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

IMAGE PROCESSING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5x2=10**
1.	**a.**	Describe the term sampling rate in two sentences.	L2	CO1	**02**
	b.	Infer the meaning of the spatial resolution in an image.	L2	CO2	**02**
	c.	Outline the two significance properties of adaptive filter.	L2	CO3	**02**
	d.	Consider the image matrix below and use connected component labeling to label the segments. [[0, 0, 0, 1, 1, 0, 0], [0, 1, 1, 1, 1, 0, 0], [0, 0, 0, 0, 1, 1, 0], [0, 0, 0, 0, 0, 0, 0], [0, 1, 1, 0, 0, 0, 0], [0, 0, 0, 0, 1, 1, 1], [0, 0, 0, 0, 0, 0, 0]]	L3	CO4	**02**
	e.	Consider the plain text "HELLO". Use the Caesar Cipher to generate the shift code of the text. (assume shift key as 3).	L3	CO5	**02**
		PART - B			**6x5=30**
2.	**a.**	Explain the types of image sampling.	L2	CO1	**05**
		OR			
	b.	Describe any five elements of the visual perception.	L2	CO1	**05**
3.	**a.**	Explain with a suitable example histogram processing.	L2	CO2	**05**
		OR			
	b.	Infer the basics of spatial filtering.	L2	CO2	**05**
4.	**a.**	What is mean filter? Explain the working principle of mean filter in image processing.	L2	CO3	**05**
		OR			
	b.	Discuss the applications and limitations of mean filter.	L2	CO3	**05**
5.	**a.**	Illustrate the steps in segmentation of a simple image matrix.	L3	CO4	**05**
		OR			
	b.	Consider the image matrix below where higher values represent brighter pixels. Employ the sobel filter with Sobel Horizontal Kernel (Gx) and Sobel Vertical Kernel (Gy) matrix values provided. Write the resulting matrices.	L3	CO4	**05**

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Image Matrix	Gx	Gy
[[10, 10, 10, 10, 10], [10, 20, 20, 20, 10], [10, 20, 30, 20, 10], [10, 20, 20, 20, 10], [10, 10, 10, 10, 10]]	[[-1, 0, 1], [-2, 0, 2], [-1, 0, 1]]	[[1, 2, 1], [0, 0, 0], [-1, -2, -1]]

6.	**a.**	Illustrate shift code methods to encode the data in images.	L3	CO5	**05**
		OR			
	b.	Demonstrate the concept of run length encoding lossless data compression in images.	L3	CO5	**05**
7.	**a.**	Illustrate the concept of texture segmentation with example.	L3	CO6	**05**
		OR			
	b.	Sketch the applications of the patterns **i.** Texture Patterns **ii.** Shape Patterns **iii.** Color Patterns **iv.** Spatial Patterns	L3	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Outline the steps involved in digital image processing.	L2	CO1	**10**
		OR			
	b.	Describe the concepts of Color models to reproduce the color in images.	L2	CO1	**10**
9.	**a.**	Explain the properties of Smoothing and Sharpening in Spatial Filtering of images with an example.	L2	CO2	**10**
		OR			
	b.	Explain color correction image enhancement technique with suitable image matrix.	L2	CO2	**10**
10.	**a.**	Describe the common methods of image restoration.	L2	CO3	**10**
		OR			
	b.	Infer the different types of image degradation models.	L2	CO3	**10**
11.	**a.**	Demonstrate the steps of Hough transformation on simple image matrix.	L3	CO4	**10**
		OR			
	b.	Consider the following 5x5 grayscale image matrix, where each number represents the intensity of a pixel. Employ the region based segmentation to find the region in an image. Choose the pixel at (0, 0) with a value of 10 as seed.	L3	CO4	**10**

		[[10, 10, 10, 50, 50], [10, 10, 10, 50, 50], [10, 10, 10, 10, 50], [70, 70, 70, 10, 10], [70, 70, 70, 10, 10]]			
12.	**a.**	Illustrate the need for data compression in images.	L3	CO5	**10**
		OR			
	b.	Consider the following characters with their corresponding frequencies. Construct the Huffman Tree and encode the characters.	L3	CO5	**10**

Character	**Frequency**
A	5
B	9
C	12
D	13
E	16
F	45

13.	**a.**	Illustrate texture and its importance in pattern recognition.	L3	CO6	**10**
		OR			
	b.	Interpret the structural texture classification methods (Gabor filter and morphological operations) with an example.	L4	CO6	**10**